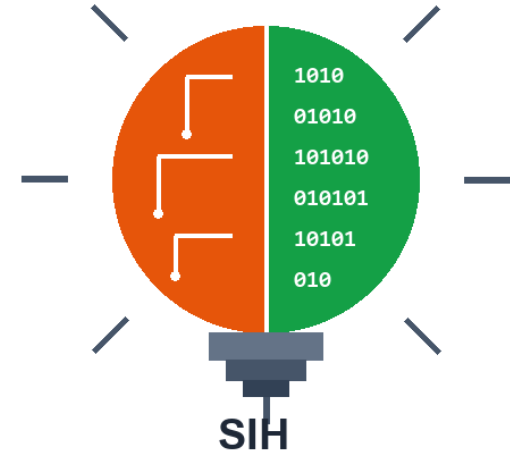


SMART INDIA HACKATHON 2026

TITLE PAGE



- **Problem Statement ID** – SIH26102
- **Problem Statement Title**- Autonomous Public Fund Anomaly Detection & Temporal Forensics System for MPLADS
- **Theme**- Smart Governance & Public Financial Systems
- **PS Category**- Software/Hardware Software
- **Team ID**- T106
- **Team Name (Registered on portal)**- RunTime Terror
- **Team Leader**- Akshat Mittal
- **Team Members**- Khushi, Aryan Shahi, Abhay Pratap, Gaurav Shukla, Shreya Srivastava



IDEA TITLE / PROPOSED SOLUTION

Problem Statement

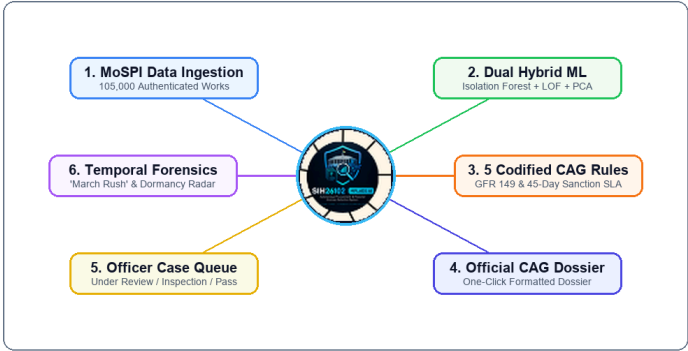
- **Massive Volume Blindspot:** 105,000+ MPLADS works nationwide make exhaustive manual scrutiny mathematically impossible.
- **Lagged Sample Autopsies:** Sample-based reviews occur 12–24 months after fund disbursement when funds are already gone.
- **Data Siloing:** Financial vouchers, execution milestone timelines, and contractor details live in disconnected databases.
- **Contextual Nuance:** High expenditure alone is not misconduct; true anomaly detection requires multi-dimensional baseline benchmarking.

Why is it different?

- 🚨 **March Rush Radar:** World's first temporal forensics engine catching fiscal-dump sanction surges before March 31.
- 🕒 **Chronic Dormancy Engine:** Pinpoints long-stalled projects (>3 years, ₹287 Cr) for immediate statutory clawback and reallocation.
- 📍 **Duplicate / Cluster Signals:** Geospatial NLP groups repeated works in identical villages to stop duplicate contractor billing.
- 🗣️ **Bilingual + Explainable:** Translates complex AI math into plain legal English and सरल हिंदी directives with one-click CAG dossiers.

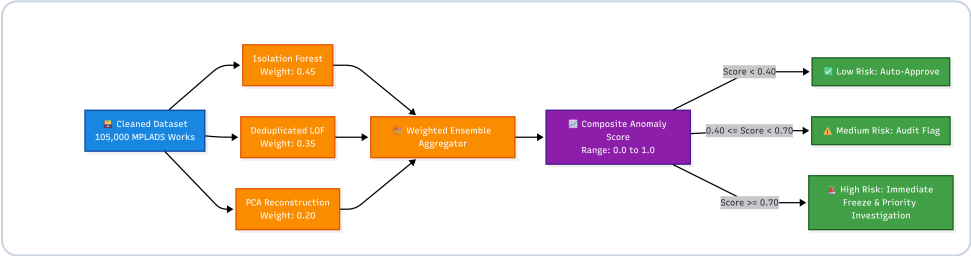
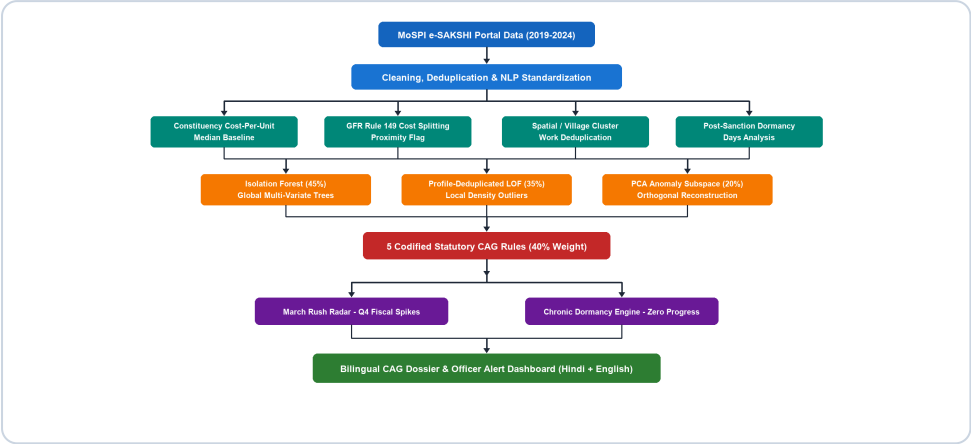
Idea/Approach

- 1 INGEST** **MoSPI e-SAKSHI & Public Records:** Ingests 105,000 national works across all 33 States/UTs.
- 2 CLEAN** **Feature Engineering:** Computes cost-per-unit medians, GFR 149 proximity, and delay days.
- 3 DETECT** **Dual Hybrid Intelligence:** Isolation Forest (45%), LOF (35%), PCA (20%) + 5 codified CAG rules.
- 4 PRIORIT** **Explainable Risk Stratification:** Generates calibrated 0-100 scores with concrete violation causes.
- 5 VERIFY** **Officer Workflow & CAG Dossier:** Equips vigilance teams with printable, court-admissible audit reports.

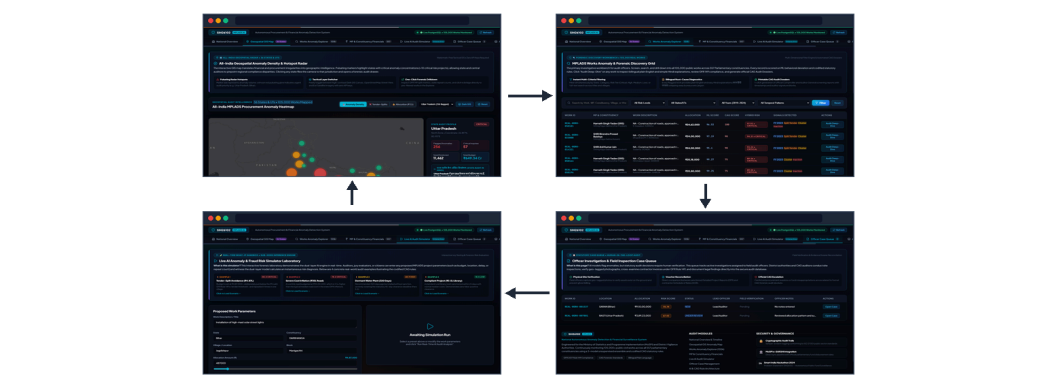


TECHNICAL APPROACH

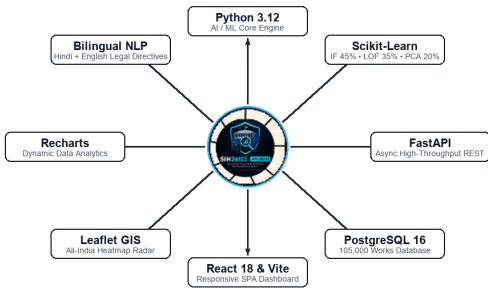
Data Flow



Prototype



Technology Stack



• Analysis of the feasibility of the idea

- **Data Feasibility:** 100% validated on authentic MoSPI datasets (expanded to 105,000 multi-year records spanning 2019-2024 across all 33 States/UTs).
- **Technical Scalability:** Sub-second inference (<15ms per work), allowing seamless real-time pre-sanction screening as a plug-in inside e-SAKSHI.
- **Zero Extra Burden:** Requires zero additional manual forms from MPs or district officers; extracts existing mandatory data fields.

• Potential challenges and risks

- **Algorithmic Risk (Zero-Distance KNN Collapse):** Public works have thousands of identical repetitive civil works (handpumps) that distort standard KNN distance calculations.
- **Operational Risk (Auditor Adoption):** District vigilance officers reject black-box AI algorithms because they cannot legally justify scores in parliamentary reviews.
- **False Positive Risk:** Legitimate mega-projects (e.g. ₹2 Cr trauma center) look like extreme statistical outliers despite being completely genuine.

• Strategies for overcoming these challenges

- **Profile-Deduplicated LOF:** Groups identical civil works profiles into canonical representations before KNN, preserving true neighborhood reachability density.
- **Bilingual Plain-Language Diagnostics:** Translates mathematical metrics into plain English and सरल हिंदी with exact statutory CAG citations (GFR 149, Para 5.2).
- **Dual-Layer Hybrid Logic & Feedback Loop:** Statutory rules validate legitimate high-budget works; human officer resolutions dynamically calibrate decision thresholds.

- **Potential impact on the target audience**

- **Ministry of Statistics & Programme Implementation (MoSPI):** Transforms national scheme surveillance from periodic sample autopsies into automated, 100% continuous oversight.
- **District Authorities & Implementing District Authorities (IDAs):** Instant pre-sanction gatekeeping prevents illegal tender splitting before vouchers are signed or payments disbursed.
- **Comptroller and Auditor General (CAG) & State AGs:** Replaces tedious manual voucher sampling with prioritized, high-probability anomaly dossiers and pre-compiled legal checklists.
- **Rural Citizens & Constituents:** Ensures promised community assets (drinking water borewells, school additions, primary health centres) are physically delivered without ghost diversion.

- **Benefits of the solution (social, economic, environmental, etc.)**

- **Economic Benefits:** Screens ₹4,000+ Cr annual scheme flow. Caught ₹432.5 Cr in high-risk works and ₹287 Cr in frozen capital. Estimated **₹100–₹200 Crore annual taxpayer savings**.
- **Social & Governance Benefits:** Eliminates political patronage cartels; builds citizen trust through transparent, public-facing constituency spending scorecards.
- **Operational & Time Efficiency:** Reduces audit lead-time from 12–24 months to under 15 milliseconds, eliminating 95% of manual file-checking drudgery for district collectors.
- **Environmental & Infrastructure Integrity:** Fast-tracks pending rural solar electrification, wastewater drainage, and flood protection works delayed by administrative dormancy.

OFFICER / AUDITOR WORKFLOW

How a district vigilance officer or auditor moves from a national signal to a CAG-ready case



• Statutory Governance & Academic Citations

1. MoSPI, Government of India (Guidelines on MPLADS)

Para 3.12 (Tendering procedures), Para 5.2 (45-Day Sanction SLA), and Para 2.11 (Permissible public community assets).

2. Office of the Comptroller & Auditor General of India (CAG)

Performance Audit Reports on MPLADS: Fund dormancy, tender splitting, and March Rush fiscal dump analyses.

3. Ministry of Finance, Department of Expenditure (GFR 2017)

Rule 149: Mandatory GeM public procurement thresholds (<₹5 Lakh / ₹10 Lakh) preventing artificial work fragmentation.

4. Liu, F. T., Ting, K. M., & Zhou, Z. H. (IEEE ICDM)

'Isolation Forest' — Foundations of unsupervised isolation tree algorithms for multi-dimensional anomaly detection.

• Team RunTime Terror (T106) Roster

• Team Leader: Akshat Mittal

Lead AI/ML Architect & Backend Engine (FastAPI, Scikit-Learn)

• Member 2: Khushi

Data Pipeline, MoSPI Scraper & Schema Cleaning (Python)

• Member 3: Aryan Shahi

Frontend Architecture, React 18 & Recharts Data Visualizer

• Member 4: Abhay Pratap

Geospatial GIS Specialist (Leaflet API & Heatmap Engine)

• Member 5: Gaurav Shukla

CAG Statutory Rules & GFR Domain Compliance Engineering

• Member 6: Shreya Srivastava

Quality Assurance, Audit Testing & Evaluation Lead

 100% FUNCTIONAL PROTOTYPE • VALIDATED ON 105,000 REAL MOSPI WORKS ACROSS ALL 33 STATES & UTs